

Neyman Design-Based Estimation: A Six-Student Example

Experimentation, A/B Testing and Causal Inference (SDA-DSC-213), SDAIA Academy. Handout for Day 2 Hours 2 and 4 (deck slides 58 to 63 and the guardrail slides) and Lab 2.

Step 1. The question and the estimand

Same setting as the sharp-null handout, now with six students: Ahmed, Bader, Dana, Reem, Sara and Tariq. Fisher asked "is there any effect at all?". Neyman asks: **how large is the average effect, and how sure are we?**

What we want, defined before any data **estimand: τ , the ATE** :

$$\tau = \text{ATE} = (1/N) \times \sum_i [Y_i(1) - Y_i(0)], N = 6$$

$Y_i(1)$ is student i 's time with the drink, $Y_i(0)$ with water. Only one is ever observed per student.

Step 2. The design

Complete randomization: exactly 3 of the 6 are picked at random for the drink ($n_1 = 3$), the rest get water ($n_0 = 3$). Arm sizes are fixed in advance; the coins only decide who goes where.

Step 3. The data **observed outcomes Y**

Student	Arm	T	Minutes to finish
Ahmed	energy drink	1	8
Bader	energy drink	1	9
Sara	energy drink	1	10
Dana	water	0	11
Reem	water	0	12
Tariq	water	0	13

Step 4. The symbols

Symbol	Meaning	Value here
N	number of students	6
n_1, n_0	students in the drink arm, in the water arm	3, 3
T_i	1 if student i got the drink, 0 if water	see table

Y_i	observed minutes for student i	see table
$\bar{y}_1, \bar{y}_0$	mean minutes in the drink arm, in the water arm	9, 12
τ (tau)	the true ATE, the estimand; unknown	?
$\hat{\tau}$ (tau hat)	the estimator $\bar{y}_1 - \bar{y}_0$; its value on our data is the estimate	-3
S_1^2, S_0^2	sample variance of minutes within each arm	1, 1
$\text{Var}(\hat{\tau}), \text{SE}$	variance and standard error of the estimate	0.667, 0.816
z	the 95 percent multiplier from the normal curve	1.96

Step 5. The estimator and the estimate

The formula applied to the data **estimator: $\hat{\tau} = \bar{y}_1 - \bar{y}_0$** is the difference in arm means. Under complete randomization it is unbiased: averaged over all possible coin flips it equals the true τ .

$$\hat{\tau} = \bar{y}_1 - \bar{y}_0$$

$$\bar{y}_1 = (8 + 9 + 10) / 3 = 9 \quad \bar{y}_0 = (11 + 12 + 13) / 3 = 12$$

$$\hat{\tau} = 9 - 12 = -3 \text{ minutes}$$

Its value on our data **estimate: $\hat{\tau} = -3$** is -3 minutes, a relative lift of $-3 / 12 = -25$ percent.

Step 6. The variance and the standard error

How much would $\hat{\tau}$ move if the coins were flipped again? First the sample variance inside each arm (denominator $n - 1$):

$$S_1^2 = \sum (Y_i - \bar{y}_1)^2 / (n_1 - 1) = [(8-9)^2 + (9-9)^2 + (10-9)^2] / 2 = (1 + 0 + 1) / 2 = 1$$

$$S_0^2 = [(11-12)^2 + (12-12)^2 + (13-12)^2] / 2 = 1$$

Then the variance of the estimate (slide 60):

$$\text{Var}(\hat{\tau}) = S_1^2 / n_1 + S_0^2 / n_0 = 1/3 + 1/3 = 0.667$$

$$\text{SE} = \sqrt{\text{Var}(\hat{\tau})} = \sqrt{0.667} = 0.816 \text{ minutes}$$

This variance is conservative. The exact formula also subtracts a term for how much the individual effects τ_i vary across students; it needs both potential outcomes, so it cannot

be estimated and is dropped. Dropping a subtraction can only make the variance larger, the safe direction.

Step 7. The confidence interval

$$95 \text{ percent CI} = \hat{\tau} \pm z \times \text{SE} = -3 \pm 1.96 \times 0.816 = -3 \pm 1.6$$

$$\text{CI} = [-4.6, -1.4] \text{ minutes}$$

The data are compatible with an average saving between 1.4 and 4.6 minutes. Zero is outside the interval.

Step 8. The test of zero average effect

Neyman's null is weaker than Fisher's: "the average effect is zero", not "no effect for anyone". Test statistic:

$$z = \hat{\tau} / \text{SE} = -3 / 0.816 = -3.67$$

On the normal curve this gives a two-sided p of about 0.0002. The catch: the normal curve is an approximation that needs many students per arm, and three is not many. Fisher's exact test on the same six students has $C(6, 3) = 20$ assignments, 2 of which show a gap this large, so the exact two-sided p is $2/20 = 0.10$. With six students, trust Fisher.

Step 9. Guardrails and the Bonferroni correction

Finishing time is the primary metric. Three guardrails were fixed before the test: test score must not fall, mistakes must not rise, self-reported jitteriness (0 to 10) must not rise. Each gets the same Neyman treatment:

Guardrail	Bad direction	$\hat{\tau}$ (drink – water)	SE	$z = \hat{\tau} / \text{SE}$	p (two-sided)
Test score (points)	lower	-1.0	1.20	-0.83	0.41
Mistakes (count)	higher	+0.7	0.40	1.75	0.08
Jitteriness (0 to 10)	higher	+1.3	0.50	2.60	0.009

Three tests at 0.05 each means the chance of at least one false alarm is above 5 percent:

$$P(\text{at least one false alarm among } m \text{ tests}) = 1 - (1 - \alpha)^m = 1 - 0.95^3 = 0.14$$

Bonferroni shares the 5 percent budget across the m tests:

Bonferroni threshold = $\alpha / m = 0.05 / 3 = 0.0167$

Equivalent form: adjusted $p = \min(1, m \times p)$, then compare with 0.05

Guardrail	p	adjusted $p = 3 \times p$	Below 0.05?	Bad direction?	Verdict
Test score	0.41	1.00	no	yes (−1.0)	OK
Mistakes	0.08	0.24	no	yes (+0.7)	OK
Jitteriness	0.009	0.027	yes	yes (+1.3)	REGRESSION

A guardrail is flagged when the adjusted p is below 0.05 and the change is in the bad direction. Jitteriness qualifies, so it is a regression. The pre-agreed rule was "no guardrail regression": HOLD, despite the three minutes saved. Fix the cause, then re-test.

Bonferroni is strict on purpose, which suits metrics that protect users; exploratory segment sweeps use the gentler Benjamini-Hochberg (Lab 2). The six-student caveat from Step 8 applies; the table is illustrative.

Step 10. The regression equivalence

Regress Y on T : $Y_i = a + b \times T_i + \text{error}$. The slope b equals $\hat{t}$ (here $b = -3$, $a = 12$).

Textbook OLS assumes equal variance in both arms; the robust HC2 standard error does not, and under complete randomization it equals the Neyman SE. Lab 2 uses `cov_type='HC3'`, a slightly more conservative cousin of HC2.

Fisher versus Neyman

	Fisher (randomization test)	Neyman (design-based estimation)
Null hypothesis	no effect for any unit (sharp)	average effect is zero (weak)
Output	an exact p -value	estimate $\hat{t}$, SE, confidence interval
Effect size	not by itself	yes, in real units
Assumptions	only the design	the design plus a large-sample normal approximation
Best when	small samples, clean significance	you need a number and an interval to decide with
Six students	$p = 2/20 = 0.10$	$\hat{t} = -3$, CI $[-4.6, -1.4]$, but too few for the approximation

Reporting the result honestly

A result is three things: the estimate in real units and as a relative lift, the interval, and the assumptions ledger (randomization unit, SE type, any adjustment). Here: "Estimated saving 3 minutes (25 percent), 95 percent interval 1.4 to 4.6; student-level complete randomization, Neyman SE; six students, so the interval rests on a weak approximation and the exact test gives $p = 0.10$."

Three things to remember

1. $\hat{\tau} = \bar{y}_1 - \bar{y}_0$ is unbiased because of the coin flips, not because of a model.
2. $\text{Var}(\hat{\tau}) = S_1^2/n_1 + S_0^2/n_0$ is conservative: the true uncertainty is this or smaller.
3. The interval needs many units per arm; with a handful, use Fisher's exact test and report Neyman's estimate as a description only.